AI Security Gateway for Detection and Prevention of Prompt Injection Attacks

A Multi-Agent Defense Matrix & Real-Time Ingress Proxy for Generative AI Applications



GROUP

Group 138

TEAM MEMBERS

Rohit Verma 23/IT/138

Rupesh Verma 23/IT/141

Utkarsh 23/IT/166

PROJECT SUPERVISOR

Dr. Akshay Mool

Assistant Professor, Department of IT

System Architecture, Literature Review & Progress (72% Completed)

End-to-End Implementation of the Multi-Agent Ingress Perimeter, Core Proxy & Operations Center

LITERATURE REVIEW & THEORETICAL FOUNDATIONS

Academic Grounding & Research Basis
Hybrid Defense Paradigm (Ji et al., 2025)
Paper: *Detection Method for Prompt Injection by Integrating Pre-trained Model and Heuristic Feature Engineering*
Core Finding: Single-model defenses fail against obfuscation. Combining heuristic feature rules with pretrained transformers maximizes F1-score and minimizes latency.
Mitigating Over-Defense (Li et al., InjecGuard 2024)
Paper: *InjecGuard: Mitigating Over-Defense in Prompt Injection Detection*
Core Finding: Direct keyword models suffer high false-positive rates on benign imperatives. Solved via multi-turn contextual deliberation and multi-agent arbitration.
Representation Circuit Breakers (Zou et al., NeurIPS 2024) & OWASP LLM01:2025
Core Finding: Establishes the necessity of independent pre-flight perimeter proxies, streaming circuit breakers, and honeypot canary tokens to prevent prompt leakage.

MULTI-AGENT INGRESS CONSENSUS ENGINE

Core Deliberation Brain (core/detection/)
Pattern Hunter (Fast Heuristics)
High-speed regex and signature engine scanning against curated threat_rules.json (<5ms execution).
Semantic Analyst (Deep NLP Transformer)
Fine-tuned DeBERTa-v3 (deberta-v3-base-prompt-injection-v2) outputting continuous adversarial semantic risk probabilities.
Behavioral Profiler (Stateful Session Tracking)
Sliding temporal window analyzing multi-turn probing escalation, boundary testing, and velocity anomalies across session states.
Judge Agent (Consensus Arbitration)
Weighted mathematical formula: (0.25 × Pattern + 0.50 × Semantic + 0.25 × Behavioral) with high-confidence override guardrails. Total deliberation latency: **<120ms**.

HIGH-PERFORMANCE PROXY & OPERATIONS CENTER

Proxy Pipeline & Monitoring (core/gateway/, static/, extension/)
Asynchronous Ingress Proxy
High-throughput FastAPI engine intercepting standard OpenAI/Gemini /v1/chat/completions. Rejects attacks with authoritative HTTP 403 Forbidden.
Active Defense & Audit Trails

- **Honeypot Canary Engine:** Dynamic canary secret injection to detect and alert on system prompt exfiltration.
- **Streaming Circuit Breaker & DLP:** Mid-stream response evaluation and regex secret scrubber.
- **SQLite Cryptographic Logging (gateway.db):** Persistent logging of prompts, risk scores, decisions, and sub-agent breakdowns.

Operations Dashboard & Client Hook

- Modern 3-column web console with live threat feeds, Chart.js analytics, and CSV export.
- Manifest V3 Chrome Extension hooking prompts on ChatGPT and Claude.

Tasks Left to be Done — Scope Expansion (Pillars I – V)

Transitioning Sentinel into a Holistic Multi-Modal AI & Cyber Threat Intelligence Platform

PILLAR I

Deep Learning Binary & Malware Analysis

MalConv (1D-CNN): Direct raw byte sequence classification with gated convolutions to identify packed PE/ELF binaries.

MalImg (2D-CNN): Byte-to-grayscale image conversion with ResNet-18 transfer learning for malware family classification.

Script Auditing: CodeBERT static model detecting obfuscated PowerShell/Bash/Python droppers.

PILLAR II

Code Security & AST Vulnerability Auditing

tree-sitter AST Traversal: Language-agnostic structural parsing preserving execution logic and data-flow graphs.

GraphCodeBERT Engine: Structural attention masking targeting CWE-89 (SQLi), CWE-78 (Command Injection), and CWE-502 (Insecure Deserialization).

Secret Leakage: High-entropy neural classification head catching exposed API tokens and private keys.

PILLAR III

Multi-Modal Vision & Steganography Defense

VLM Jailbreak Defense: Dual pipeline (OCR + CLIP cross-modal cosine similarity) stopping visual prompt injections.

Steganography Classifier: Spatial Rich Model (SRM) with residual high-pass filters detecting LSB and FGSM/PGD noise.

PILLAR IV

Agentic Runtime Guardrails & Tool Sandboxing

Tool-Call Inspection: Schema validation, shell command filters, and DistilBERT model preventing injection via tool arguments.

Memory Poisoning Defense: Vector embedding distance checks and secondary LLM meta-instruction auditing for RAG memory.

PILLAR V

Autonomous Red-Teaming & Compliance Auditing

Attacker-Defender Loop: Closed-loop RL attacker agent iteratively testing multi-turn obfuscated jailbreaks against the gateway.

Automated Auditing: Automated evaluation against OWASP Top 10 for LLMs and MITRE ATLAS, generating cryptographic audit reports.

Expected Timeline & Project Completion Milestones

End-to-End Project Lifecycle: Completed Deliverables (72%) vs. Remaining Scope (28%)

72% completed — current state

Milestone 1 — Problem Formulation & Threat Modeling

- In-depth vulnerability research based on OWASP Top 10 for LLMs (LLM01:2025).
- Formalization of the perimeter defense model to mitigate direct and indirect prompt injection attacks.

Milestone 2 — Multi-Agent Ingress Detection Engine

- Implemented high-speed heuristic scanner (**Pattern Hunter**) with curated threat_rules.json.
- Integrated fine-tuned Transformer (**Semantic Analyst** — DeBERTa-v3) for continuous adversarial intent scoring.
- Implemented stateful session tracking (**Behavioral Profiler**) for multi-turn probing and velocity anomalies.
- Deployed weighted consensus arbitration (**Judge Agent**) with high-confidence semantic overrides.

Milestone 3 — Asynchronous Proxy, Telemetry & Operations Center

- Built high-performance async FastAPI proxy intercepting /v1/chat/completions (<120ms latency).
- Deployed honeypot **Canary Engine**, streaming circuit breaker, and Response DLP secret scrubber.
- Built real-time interactive **Sentinel SOC Console** with persistent SQLite audit logging and Chart.js analytics.

28% remaining

Milestone 4 — Deep Learning Binary & Source-Code Security Auditing

- **Pillar I:** Train MalConv 1D-CNN on raw byte sequences and MalImg 2D-CNN byte-to-image ResNet-18 family classifier. CodeBERT static script analyzer.
- **Pillar II:** tree-sitter AST parsing and GraphCodeBERT with structural attention masking for CWE-89 (SQLi), CWE-78 (Command Injection), and CWE-502 (Insecure Deserialization). High-entropy secret leakage head.

Milestone 5 — Multi-Modal Vision Defense & Agentic Runtime Guardrails

- **Pillar III:** OCR + CLIP cross-modal cosine similarity visual prompt rejection and Spatial Rich Model (SRM) steganography filter (excluding audio).
- **Pillar IV:** Tool-call argument inspection (schema validation + DistilBERT injection filter) and vector memory poisoning protection.

Milestone 6 — Autonomous Red-Teaming, Hardening & Final Viva

- **Pillar V:** Closed-loop Attacker-Defender RL loop iterating multi-turn obfuscated jailbreaks; automated OWASP/MITRE ATLAS benchmark evaluator.
- Triton/ONNX inference acceleration, ISO/NIST compliance audit reporting, and final B.Tech Minor Project dissertation defense.